



Ultrasound Cavitation/Microbubble Detection and Medical Applications

Zahra Izadifar¹ · Paul Babyn² · Dean Chapman³

Received: 26 June 2017 / Accepted: 2 February 2018 / Published online: 11 April 2018
© Taiwanese Society of Biomedical Engineering 2018

Abstract

Over the past decades, different techniques have been investigated for detecting microbubbles. The purpose of this work is to review the state-of-the-art of medical microbubble detection along with therapeutic, monitoring, and diagnostic applications. The presence of microbubbles in the human body can be induced either through cavitation or exogenous introduction of bubbles. One of the effects of ultrasound is cavitation, or microbubble formation and collapse. Cavitation produces high pressures and temperatures, and microbubble expansion and then collapse close to cells can lead to cellular damage or hemorrhage in biological tissues. Cavitation is, in most cases, an undesired event in clinical diagnostic imaging. Considering that cavitation microbubble formation is largely unpredictable, ultrasound imaging may present a rare or yet unknown risk, particularly to fetuses and embryos. Although most therapeutic ultrasound modalities work based on physical and thermal effects of cavitation, the safety of treatment strongly depends on accurate knowledge of the location of the cavitation inception point. Cavitation detection is an important factor with respect to improving the safety of ultrasound imaging and therapy. It is essential to recognize the existence and location of cavitation inception points. In addition, the use of encapsulated microbubbles as contrast agents for diagnostic imaging, as vehicles for local drug or gene delivery, and as tools for microbubble and ultrasound therapy in thrombolysis has increased the demand for an accurate deep tissue microbubble detection technique. There have been many attempts to detect cavitation bubbles, but each contains its own limitations. There is no doubt that continuous discoveries and developments in microbubble detection modalities will lead to safer and more efficient therapeutic and diagnostic equipment.

Keywords Ultrasound · Detection technique · Cavitation bubble · Microbubble contrast agent

Abbreviations

ABI Analyser based imaging
ACD Active cavitation detection

BMIT Biomedical imaging and therapy
CLS Canadian light source
DPCD Dual passive cavitation detection
ESR Electron spin resonance
ESWL Extracorporeal shock wave lithotripsy
HIFU High-intensity focused ultrasound
MI Medical index
MRI Magnetic resonance imaging
PCD Passive cavitation detection
PCI Phase contrast imaging

✉ Zahra Izadifar
zai206@mail.usask.ca

Paul Babyn
paul.babyn@saskhealthauthority.ca

Dean Chapman
dean.chapman@usask.ca

¹ Division of Biomedical Engineering, College of Engineering, University of Saskatchewan, 57 Campus Drive, Saskatoon, SK S7N 5A9, Canada

² Department of Medical Imaging, University of Saskatchewan and Saskatoon Health Region, Royal University Hospital, 103 Hospital Drive, Saskatoon, SK S7N 0W8, Canada

³ Anatomy and Cell Biology, University of Saskatchewan, 3B34 Health Sciences Building, 107 Wiggins Road, Saskatoon, SK S7N 5E5, Canada

1 Introduction

The purpose of this work is to review the various modalities that have been used for detection of microbubbles/cavitation bubbles, and their relative advantages and disadvantages.

It has been shown that when the negative pressure of the acoustic wave propagated through a liquid is strong

enough (~ 10 MPa in shockwave lithotripsy) to rip the fluid apart, then bubbles are generated [1]. Cavitation is defined as the formation of bubbles (cavities) in a liquid. The term cavitation process is used to describe the process in which vapor- or gas-filled cavities undergo growth and implosion or resonance in a liquid host medium on exposure to acoustic radiation. Cavitation event can be either inertial or stable cavitation process. More detail about cavitation phenomena are given in next section of this article. The fundamental physics of the cavitation process in human tissues during clinical ultrasound applications still requires further study.

Multiple physical effects are associated with the growth and violent collapse of cavitation bubbles, including direct physical phenomena such as luminescence, free radical formation, very high pressure shock wave emissions, shear stress, and high-speed microjet production [2]. In ultrasound diagnostic imaging, cavitation can be an undesired and unwanted phenomenon. Although some therapeutic modalities work based on cavitation, it can also be considered an undesired bioeffect (e.g., giving rise to hemorrhaging) in other therapeutic modalities (e.g., lithotripsy).

There has been interest in the use of higher frequencies (up to 10 MHz) and higher acoustic power (up to 690 W/cm², spatial peak, pulse average) for improved imaging resolution [3]. In addition, more recent interest in three-dimensional fetal images as souvenirs in obstetric ultrasonography raises concern with respect to potential bioeffects. It has been reported that ultrasound transmitted into a tissue may have a rarefactional pressure of for example 2 MPa that represent a negative tensile 20 times atmospheric pressure (i.e., 0.1 MPa) that can initiate cavitation activity in tissue. It has been interpreted that as a mechanical bioeffect, cavitation or directly generated pulsation of pre-existing gas bodies, such as in lung, intestine, or exogenous bubbles, local tissue injury in the immediate vicinity of the cavitation activity, including cell death and hemorrhage of blood vessels can occur [4].

Because cavitation plays a significant role in therapeutic modalities such as extracorporeal shock wave lithotripsy (ESWL), intracorporeal lithotripsy, and the ultrasonic cavitation devices used in surgery to dissect or fragment tissues, the safe application of these modalities largely depends on precisely locating the cavitation microbubbles generated by these devices.

Microbubbles can also be introduced into human tissues by injection and are used as microbubble contrast agents for ultrasound diagnostic imaging or as vehicles for drug or gene delivery. Considering that the entrance of medical microbubbles (e.g., ultrasound contrast agents) into the human body is followed by ultrasound radiation, the growth and violent collapse of microbubbles can have the same or similar effect as cavitation on cells, blood vessels, or tissues. Therefore, cavitation bubbles and ultrasound contrast agents

should both be further studied in the body to provide safety assurance and minimize the side effects of clinical ultrasound modalities. However, to accurately assess and control cavitation, it is first essential to determine the existence of microbubbles and locate any cavitation inception points.

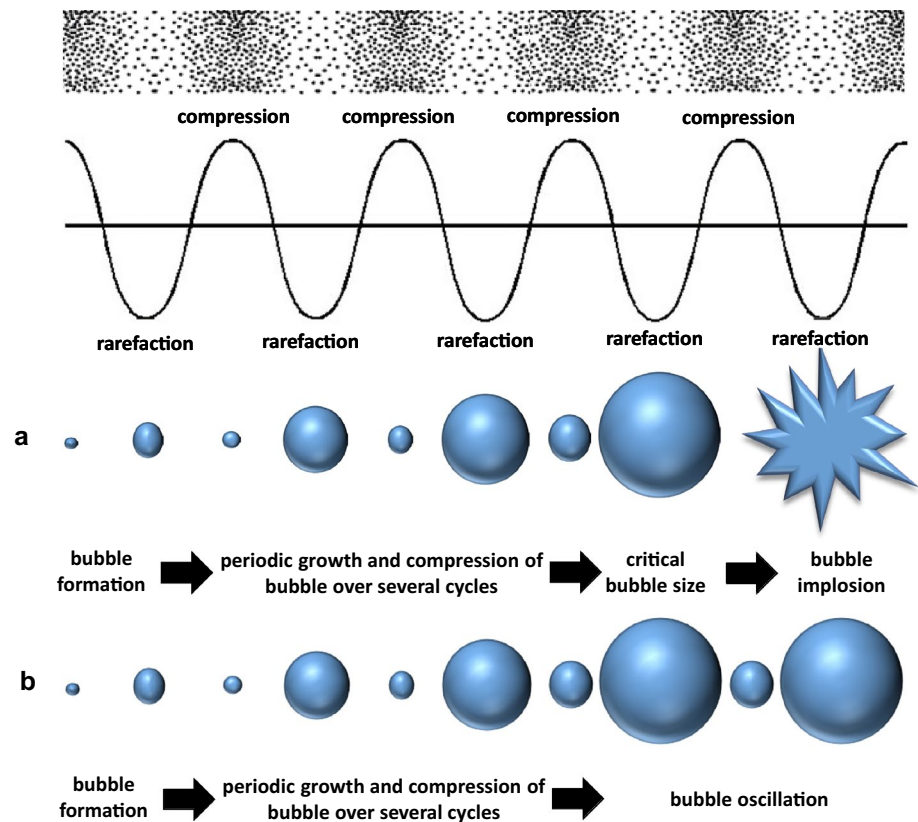
This review briefly describes the fundamentals and clinical applications of cavitation phenomena and ultrasound microbubble contrast agents. The biological effects of cavitation/microbubbles and concerns regarding their adverse effects on tissue and organs in the human body are discussed. This is followed by a review of the various modalities that have been used for detection of microbubbles/cavitation bubbles, and their relative advantages and disadvantages.

1.1 Understanding Cavitation Phenomena

The physics of cavitation process in fluid is relatively well described. The possible mechanism that could explain the induction of cavitation bubble is: when an ultrasound wave is applied to a liquid the molecular structure of the medium goes through alternating expansion and compression cycles. As expansion (rarefaction) and compression cycles travel through a medium, rarefaction pulls molecules apart and compression pushes them together. If the ultrasound wave is strong enough, the expansion cycle can overcome intermolecular binding forces and lead to a sudden pressure drop and creation of bubbles of gaseous substances in the liquid. These bubbles grow in size with the ensuing expansion cycles of ultrasound until they reach an unstable size and then violently collapse (Fig. 1). This physical phenomenon is called acoustic cavitation.

Based on the calculated results from a developed crevice model of nucleation in water, cavitation bubbles are typically initiated at a micron size in the fluid [5] but, depending on the frequency, power, and environment, their size will vary. For example, in a lithotripter shock wave the bubble can grow from a 1 μ m radius to a ~ 1 mm radius over a period of 150 μ s; it then takes a further 150 μ s for the bubble to collapse [1]. These radius values of a spherical bubble subject to a lithotripter pulse were predicted as a function of time in water with negative pressure of -4.2 MPa and positive pressure of 34.5 MPa [1]. The size of cavitation bubbles primarily depends on the ultrasonic frequency. At a given ultrasonic power, higher frequencies produce smaller cavitation bubbles and lower frequencies generate bigger cavitation bubbles [6]. This is partially due to the fact that the increased wavelength at lower frequencies allows more time and space for the bubbles to grow to larger sizes before they implode. The energy released in the zone as the bubbles implode directly relates to the bubble size. Larger bubbles require more energy to grow and, in turn, a larger amount of energy will be released when they implode [6]. Decreasing the frequency enhances the likelihood of cavitation activation [4].

Fig. 1 Schematic image of acoustic **a** inertial and **b** stable cavitation process



Based on the conservation of energy, at a constant frequency a higher ultrasonic power will lead to a larger number of cavitation bubbles [6].

Two types of cavitation bubbles form based on the behavior of microbubbles within a medium. When a bubble forms in a liquid medium, it grows until it reaches a critical size known as its resonance size, which depends on the frequency of the applied sound field. If the microbubble reaches its resonance size and then becomes unstable and collapses violently within a single cycle or over a small number of acoustic cycles, it is called inertial cavitation (or transient cavitation). If, however, the microbubble does not violently collapse and instead oscillates for many cycles at, or around, its resonance size, it is called stable cavitation (or non-inertial cavitation) [7]. In stable cavitation, the ultrasound pressure acts as a driving force that changes the bubble size and shape; the bubble behaves as an oscillator with a stiffness provided by the gas within the bubble and inertia provided by the liquid surrounding the bubble that moves with the bubble wall. Generally, higher acoustic pressures lead to inertial cavitation and lower acoustic pressures result in stable cavitation [8].

The whole inertial cavitation process happens within about 400 μ s. Normally, cavitation occurs at pre-existing weak points in the fluid, such as gas-filled crevices in suspended material or transient bubbles remaining from

previous cavitation events [9]. In successful cavitation processes, the cavities can grow and collapse violently with the release of energy in the form of an acoustic shock wave, temperature, pressure, and visible light [10]. Based on the hot spot theory, the estimated rapid adiabatic compression of gases and vapours within the bubbles or cavities produces hot spots with extremely high temperatures and pressures that approach 5000 $^{\circ}$ C and 1000 atm, respectively, during bubble collapse [11].

Because the size of the bubbles is very small relative to the total liquid volume, the heat produced by bubble collapse is rapidly dissipated with generally no appreciable change in overall environmental conditions [9]. The cooling following the implosion of a cavitation bubble is estimated to be in the region of 10 billion $^{\circ}$ C/s [9]. The bubble collapse can cause emission of light and also gas temperatures that can reach over 1000 $^{\circ}$ C [9]. The inertial cavitation process can potentially create free radicals or give rise to emission of light, which is known as 'sonoluminescence'.

When the acoustic cavitation bubble collapse is close to or on a solid surface, the dynamics of collapse change dramatically. In the absence of any close physical boundary, the bubble retains its spherical shape during collapse because its surroundings are uniform. However, close to a physical boundary, it can only collapse asymmetrically because the solid surface provides resistance to liquid flow. It has been

reported that this asymmetric collapse produces high-speed liquid jets driving into the surface of the solid at speeds close to 111 m/s [9]. The impact of the jets on the surface is very strong and can result in serious damage to impact zones and cause surface pitting (erosion). After its collapse, the bubble might fragment or may repeat its growth/collapse cycle. The probability of inertial cavitation occurrence for a particular bubble nucleus depends on the acoustic pressure amplitude, the acoustic frequency, and the bubble radius. The influence of the physical characteristics of fluid viscosity and surface tension on the dynamics of cavitation bubbles was studied by Plesset, who derived the Rayleigh-Plesset equation in 1949 [12]. In any cloud of bubbles in the medium within the same ultrasound field, both inertial and non-inertial cavitation may occur simultaneously. However, when the bubbles are small the surface tension prevents initial growth and therefore the bubbles do not grow. When the cavitation nuclei are large, they can grow initially but do not collapse to generate sufficiently high temperatures. Therefore, for a specific frequency and nucleus radius a threshold pressure is required to initiate inertial cavitation. Apfel and Holland studied the theoretical threshold acoustic pressure for inertial cavitation in blood and water and reported that if the square of peak negative pressure over frequency (p^2/f) be two to three orders of magnitude above 0.5, then there is a potential for bubble activity and the bioeffects definitely occurs [13]. If the bubbles are close to an elastic medium, their motion along the surface (shear motion) can lead to an increase in the heating of the medium (shear loss) [14]. When the bubble is pulsing, the inhomogeneous periodic field around the bubble can produce a small steady flow of fluid known as microstreaming [15]. Extremely high shear stresses near the bubble surface will be produced because of the variation of this flow with distance from the bubble. Cell membrane destruction and temporary alterations in permeability have been associated with this high shear stress.

1.2 Effect of Cavitation/Microbubbles on Body Tissues and Fluids

The presence of microbubbles in the body can result from either exogenous introduction of microbubbles into the body or endogenous generation of microbubbles in the body as a result of ultrasound propagation. The properties of endogenous microbubbles generated within the area of an ultrasound beam will depend on the ultrasound wave's properties. Cavitation microbubbles or endogenous bubbles can be intentionally produced in the body for clinical applications such as lithotripsy, as explained later in this review.

Ultrasound microbubble contrast agents (medical microbubbles) are exogenous microbubbles generated outside of the body and then injected into the vasculature for different medical applications. The presence of either endogenous

or exogenous microbubbles can have multiple thermal and biological effects.

At high ultrasound pressure [mechanical index (MI) > 1.0], microbubbles are forced to expand and then are compressed, which results in their destruction. Destruction of these microbubbles close to a living cell leads to permeability of cell membranes [16]. The destruction of microbubbles with ultrasound has caused rupture of microvessels with extravasation of red blood cells [17]. The impact of the destruction or oscillation of microbubbles on cell membrane causes reparable submicron pore formation [18]. The dynamic features of a population of microbubbles have a time scale relevant to pulsed ultrasound exposure [19]. The microbubble cavitation physics is in timescale of $\sim \mu\text{s}$ that is coupled with $\sim \text{ms}$ timescale of cell permeability dynamics and $\sim \text{second to minutes}$ timescale of the resulting cellular biophysics effects [20]. It has been reported that there is a direct correlation between microsecond-scale microbubble oscillation physics and second-to-minute-scale macromolecular uptake and cell membrane reorganization. For example, immediate generated pore membrane through both apical and basal cell membrane layers reseal along their lateral area in approximately less than 2 min [20]. As a mechanical bioeffect, cavitation activation can cause local tissue injury in the immediate vicinity of the cavitation activity, including cell death and hemorrhage of blood vessels [4]. For example, cavitation-mediated tissue damage in shockwave lithotripsy has been reported [21]. Large surface hemorrhaging over both the targeted and contralateral kidneys along with hemorrhaging of the spleen, intestine, and peritoneum have been observed in small animal or in vitro studies [21]. Tissue histology has revealed vascular rupture in the kidney, necrosis of the walls of the intralobular arteries and veins, and diffuse damage to the targeted kidney tissue that may be due to cavitation [21].

The exact mechanism by which microbubbles enhance the permeability of cells is still unresolved [16]. When an ultrasound wave is applied to tissues, it creates cavitation bubbles in the fluid that, along with the cavitation process, result in an incremental increase in cell permeability. In fact, the cavitation process on its own increases cell permeability [22]. However, the presence of additional microbubbles along with the high acoustic pressure due to ultrasound exposure have additional effects on cell permeability [23]. Cavitation in body tissues or blood sets fluid in motion and makes small shock waves that give rise to microstreaming along the endothelial cells [16]. High-energy microstreams or microjets produced by cavitation bubbles during the cavitation process cause shear stress on the membrane of an endothelial cell [24], which increases the cell permeability. This increase is probably due to transient holes in the plasma and nuclear membrane [16]. Other proposed mechanisms for cell permeability increase include the generation of reactive

oxygen species in endothelial cells during the cavitation process [25], an increase in intracellular radical production (which is associated with cell killing in vitro and endothelial cell layers), a rise in temperature (which can influence the fluidity of phospholipid bilayer membranes), and endocytosis or phagocytosis (which may activate membrane transport mechanisms) [16]. Bioeffects are mainly influenced by factors such as the choice of ultrasound frequency and its amplitude, sonoporation and presence of exogenous microbubbles that can intensify the ultrasound effect on the cell membrane which need further investigation.

2 Ultrasound Microbubble Contrast Agents Application

2.1 Diagnosis and Monitoring

Many of the ongoing improvements in sonographic diagnosis have been related to the entry of ultrasound microbubble contrast agents (medical microbubbles). Small exogenous gas bubbles are used to increase ultrasound contrast. These medical microbubbles are used to increase the reflectivity of perfused tissue in cardiology and radiology applications [26].

Microbubbles used in echocardiography as contrast agents are small gas-filled microspheres with specific acoustic properties. First generation microbubbles were room air microspheres [27] that disappear a few seconds after intravenous administration [28, 29]. Second generation microbubble contrast agents are filled with a heavy molecular weight gas such as sulphur hexafluoride and stabilized with a thin shell such as sonicated albumin and (phosphor)lipids to improve survival and stability of microbubbles under high pressure [27]. Second generation microbubbles have a much smaller diameter (about 2.5 μm) than first generation microbubbles to ease their passage through the pulmonary capillary bed [16]. Diagnostic imaging with contrast agents is improved when an acoustic pressure higher than 0.05 is used (mechanical index, $\text{MI} > 0.05$). An MI higher than 0.05 causes emission of non-linear harmonic signals at multiples of the transmitted frequency [30]. This improves the signal to noise ratio and creates an acoustic impedance mismatch between body tissues and fluids containing microbubbles, which makes these microbubbles beneficial in diagnostic ultrasound imaging [31].

2.2 Ultrasound Drug Delivery

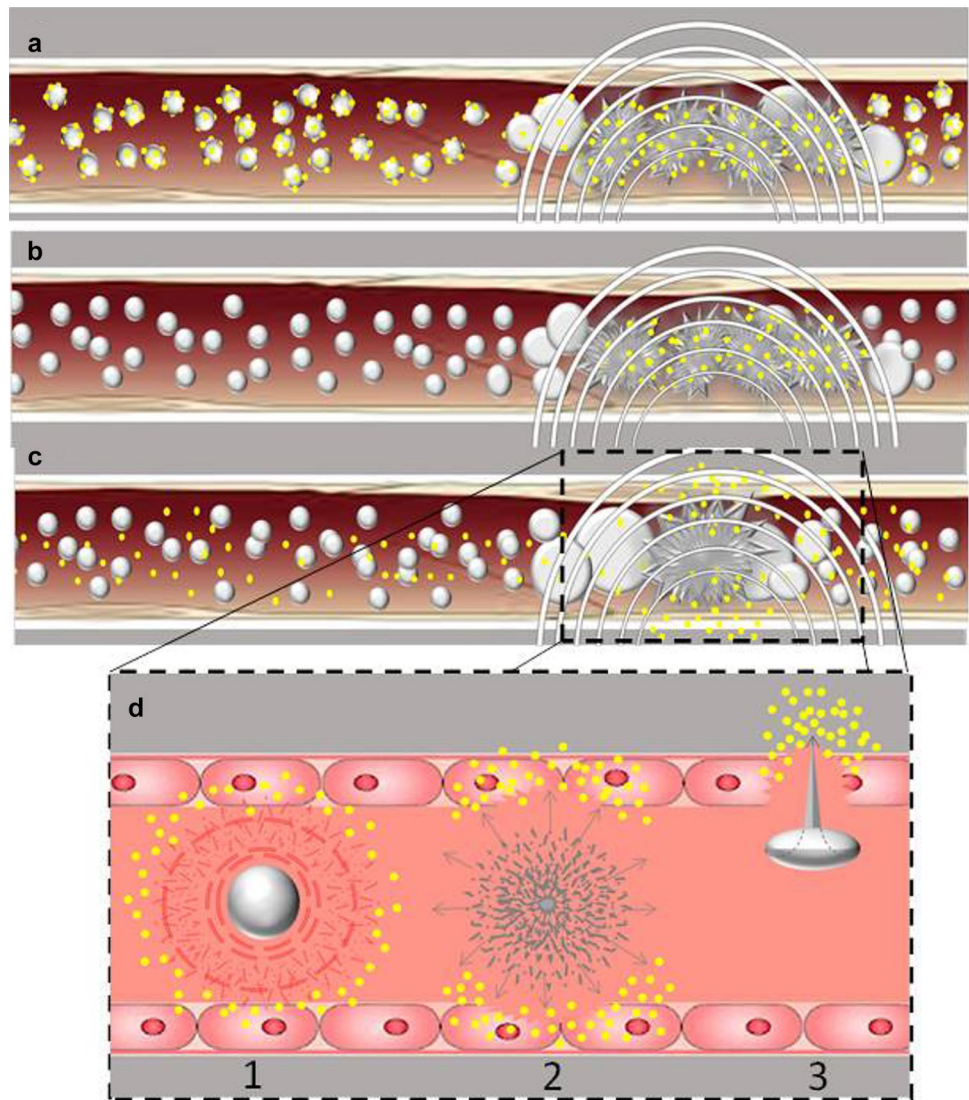
Cell membranes often prohibit large molecules (e.g., drugs and genes) from entering cells. The mechanical force of focused ultrasound through endogenous microbubbles of stable (or non-inertial) cavitation can improve cell membrane

permeability and give rise to absorption of drugs or genes. This effect, known as sonoporation, creates pores in cell membranes via stable cavitation and allows a greater volume of compounds to enter the cell [32]. In addition, the acoustic streaming induced by stable cavitation gives rise to the flow of fluid in the local environment of the cell. This helps open pores in the cell membrane and also directs the molecules of the drug toward the cells [33, 34]. The blood–brain barrier (BBB) that is formed by brain endothelial cells prevents the brain uptake of most pharmaceuticals. The shear stress induced by burst-mode focused ultrasound induced cavitation microbubbles in the vasculature of the brain can temporarily disrupt the tight junction of BBB and increase the permeability of therapeutic agent to the brain [35, 36]. Also, it has been reported that focused ultrasound BBB opening causes brain neuromodulation through ultrasound-induced mechanical stress [36]. The neuromodulation effect of ultrasound differ across cortical areas and can, for example, increase neuronal activity in the motor cortex and suppress activity in the visual cortex [37]. The exposure condition of focused ultrasound plays an important role in brain neuromodulation. Brain stimulation techniques are done with low intensity focused ultrasound and major histological analysis confirms no side effects on surrounding tissues of the region of interest that concluded that no cavitation may occur [38, 39]. But low-intensity burst type focused ultrasound combined with exogenous microbubbles have been shown to open BBB [40, 41] and induce neuromodulation at specific exposure conditions [36].

On the other hand, several properties of exogenous medical microbubbles make them promising tools for delivery of drugs and genes to living cells [42–48]. Ultrasound contrast agents combined with focused ultrasound is uniquely suitable for localized drug delivery. Many diseases such as cancer, inflammatory diseases, or thrombo-embolic processes may require high concentrations of certain drugs. However, high drug concentrations can have toxic side effects on the rest of body. In addition, systemic drug delivery requires plasma concentrations to be within the therapeutic range, but this is limited by the occurrence of potential side effects. The presence of circulating exogenous microbubbles with focused ultrasound can be used for drug delivery. Low intensity focused ultrasound with exogenous microbubbles can temporarily disrupt BBB and ease drug delivery for treatment of brain diseases such as brain tumor [40]. It is non-invasive, reversible, localized, with high accuracy if delivered at a proper intensity of ultrasound [49], however the optimum intensity without any probable injury to the brain remains to be determined [50].

In more specific drug delivery, microbubbles can be coated with a drug by attaching it to a ligand on the outside of the drug-laden microbubbles (Fig. 2a). Focused ultrasound is then applied locally at the targeted area, resulting

Fig. 2 Illustration of variety of ways drug delivery can be enhanced by utilizing microbubbles. **a** Microbubbles with drug-laden external membranes are freely circulated in the vessel and bind in the target region, then are ruptured by ultrasound and the drug payload liberated in the target area; **b** Microbubbles carrying drugs inside of them are circulated in the vessel and destroyed by ultrasound and the transported substances thus released into the surrounding targeted tissue; **c** Free circulation of drug particles (yellow circles) along with ultrasound microbubbles (grey circles) in vessels, and the effect of ultrasound on growth and burst of microbubbles results in extravasation of drug; and **d** Schematic of various modes by which microbubbles increase the cell permeability: 1 non inertial cavitation when a bubble behaves as an oscillator and microstreaming around bubble increases the permeability; 2 collapse of a gas bubble and emission of a shock wave; and 3 asymmetric bubble collapse close to a cell wall producing a high speed liquid jet that pierces and ruptures the cell



in selective delivery of the drug in the body [51]. Alternatively, drugs may be incorporated into the microbubbles and released in the region of interest by rupturing the microbubbles with localized focused ultrasound (Fig. 2b) [52, 53]. Another approach is to systematically and simultaneously inject the drug and an ultrasound contrast agent and then apply focused ultrasound to the target region (Fig. 2c). The oscillation and destruction of the bubbles in small vessels alters the vessel walls and results in extravasation of the drug [54, 55]. As shown in Fig. 2d, the permeability of cells during drug delivery can be increased (1) through stable cavitation when the bubble behaves as an oscillator and microstreaming around the bubble increases the permeability (Fig. 2d-1), (2) through collapse of the gas bubble and emission of a shock wave (Fig. 2d-2), or (3) through asymmetric bubble collapse close to the cell wall producing a liquid jet that pierces and ruptures the cell (Fig. 2d-3). Recently, combine ultrasound, microbubble, and chemotherapy has been

studied in human clinical trials for treatment of inoperable pancreatic cancer [56, 57]. In these studies, Gemcitabine (a chemotherapy medication) was infused intravenously and patients were treated with ultrasound followed by microbubble (SonoVue®) injection intravenously during ultrasound treatment. The results showed decrease in tumor growth and diameter, and increase in patients' survival rate and tolerance for more number of gemcitabine cycles (compared with historical controls). This combined treatment may improve the clinical efficacy of gemcitabine and prolong the quality of life in patients.

Drug delivery via drug-loaded microbubbles that are ruptured with localized ultrasound can result in local drug release in higher concentrations than can be achieved with systemic administration. In addition, the drug will be pushed directly into the cell given the higher cell permeability in this technique [16]. Drug and gene delivery with microbubbles is a promising technique; however, studies have shown

that sonoporation as a tool for local delivery of therapeutic macromolecules creates reparable/resealing submicron pores through both layers of cellular membrane, and gaps between confluent cells [18, 20]. Adverse effects such as a rise in blood temperature or hemolysis, and a wide range of non-thermal mechanical bio-effects at the cellular level such as cell killing and apoptosis, or deformation of microvessel walls, and submicron pore formation are possible side effects that have been a cause of concern in several studies [16, 58–60].

2.3 Gene Therapy with Microbubbles

Gene therapy may play a significant role in the treatment of several (cardiovascular) diseases in the near future. In gene therapy, first the gene needs to be delivered to the tissue and taken up by the cells. Then, it should be incorporated into the genome in the nucleus of the cell without being digested. The direct intravascular injection of genes is not possible as the DNA consists of large molecules and results in the removal of the gene from the blood. In addition, the gene is too long to enter the cell passively [16]. Several studies have shown that ultrasound in combination with microbubbles can be a safe method of gene delivery to cells [22, 42, 61–63].

2.4 Targeted Microbubbles in Diagnostic Applications

Considering that microbubbles in current use are comparatively stable and circulate throughout the body, it is possible that the contents of the microbubbles may be deposited in undesired tissues. To avoid this problem, microbubbles can be combined with ligands and receptors that are incorporated in the bubble shell, creating “targeted microbubbles” that target specific tissues [16]. These targeted microbubbles can also be useful for active attachment of microbubbles to target tissues and provide further potential for diagnosis or localized drug delivery to targeted lesions for therapy. This form of molecular imaging aided by targeted microbubbles, known as targeted contrast-enhanced ultrasound, facilitates diagnosis of cancer, diseased or abnormal tissues. Targeted contrast-enhanced imaging in first human trial in prostate cancer showed feasibility, safety and promising results in multiple preclinical models regarding cancer imaging [64]. Another first-in-human clinical trial in patients with breast and ovarian lesions reported a well signal match between targeted contrast-enhanced ultrasound imaging and immunohistochemistry (as the gold standard) in addition to clinically safe and feasible [65].

Targeted microbubbles have been described in several studies where they have targeted different types of tissues and processes, including endothelial cells (for diagnosis of

preclinical atherosclerosis) [66], non-invasive identification of acute cardiac transplant rejection in mice [67], inflamed tissue and angiogenesis [68] (for ultrasound imaging of inflamed tissue [69, 70] such as post-ischemic myocardial inflammation [71]), and vascular clots or thrombi (visualization of vascular clots associated with cardiac vascular disease such as stroke and myocardial infarction, which can be used for identification and cure of vascular clots in humans) [72]. Ultrasound is known to improve clot lysis in blood vessels, and ultrasound in combination with targeted microbubbles loaded with thrombolytic agents can further drive this effect by cavitation [16].

The application of ultrasound at low frequencies (from 26 kHz to 10.03 MHz) and high power in the presence of absence of fibrinolytic therapy has been shown to increase clot lysis in both in vitro and in vivo studies [73–80].

The presence of microbubbles lowers the energy needed for cavitation. Destruction of microbubbles by high power ultrasound creates powerful microjets that accelerate the effect of ultrasound in thrombus dissolution compared to ultrasound alone [16]. These assessments have been reviewed in [16]. The effect of the cavitation process on the surface of the vascular clot plays an important role in micro fragmentation, which can make the clot more susceptible to fibrinolytic agents [81]. The influence of different types of microbubbles and different ultrasound operating conditions (170 kHz and 0.5 W/cm [82], 10 MHz and 0.5–1.0 W/cm [83], 20 kHz and 1.5 W/cm [84], 20 kHz and 40 W/cm [85], 10 MHz and 1.02 W/cm [86]) have all shown significant improvement on thrombolysis. In a recent human trial in an unselected acute ischemic stroke population, neurological improvement and favorable functional outcome were observed using contrast-enhanced sonothrombolysis [87]. In this study that either contrast-enhanced sonothrombolysis (CEST) or sham CEST was used, application of ultrasound wave was performed at 2 MHz for 1 h. The Magnetic resonance imaging and angiography at 24 h after procedure showed neurological improvement. However, no statistically significant clinical effect was observed in the reported prematurely stopped clinical trial. Endothelial cell injury of microvessels, which can be a potential clinical danger, has been reported by Kobayashi et al. [88]. Diagnostic ultrasound imaging commonly works at frequencies between 2 and 15 MHz [89]. Most cavitation processes occur at lower frequencies and higher powers, and thrombolysis by ultrasound would require such operating conditions. Considering the bioeffects of ultrasound at low frequency and high power, the safety of all of these techniques requires further study; microbubble detection techniques, especially for future applications in humans, are also needed. At the cellular level, the interaction of fluorescent-labeled microbubbles and myocardial or endothelial cells under high ultrasound pressure can be precisely studied using live-cell imaging

techniques (such as multi-dimensional digital imaging microscopy) [16] and two-photon microscopy [90]. But to control the exact area that microbubbles deposit the gene or drug in the body or are interacting with live cells deep in tissue, an imaging technique capable of detecting microbubbles deep in tissue (such as the human body) without disturbing the microbubbles is required.

3 Detection and Imaging of Cavitation and Microbubbles

Over the past decades, different techniques have been investigated for detecting and mapping microbubbles. These techniques mainly include optical, acoustic, and scattering methods. Optical methods include high-speed photography, sonoluminescence, and sonochemiluminescence, and acoustic methods include active cavitation detection (ACD) and passive cavitation detection (PCD), and the Doppler method. Scattering methods include laser and synchrotron based X-ray methods. Each of these techniques has specific strengths and weaknesses for detection and mapping microbubbles. These signatures can be used to detect the microbubbles through physical and chemical techniques. Most published studies of visualization and detection of microbubbles come from *in vitro* experiments. There have been few attempts to detect cavitations in animals such as a 30 MHz active cavitation detection from a diagnostic ultrasound pulses in the rat lung, and an active detection using echo on B-mode ultrasound, plus passive detection using coincident signals on orthogonal receivers to detect cavitation in pig kidney during shock wave lithotripsy [91, 92]. One challenge for *in vivo* detection of the cavitation microbubbles is attributed to the high attenuation and inhomogeneous nature of tissues which affect the sound wave propagation and consequently the behavior of the ultrasound-induced microbubbles *in situ*. A couple of techniques can also measure either exogenous microbubble or endogenous cavitation bubbles. In the following section, the different modalities that have been used to study microbubbles are briefly reviewed.

3.1 Optical Methods

3.1.1 Detection of Physical and Chemical Responses

The presence of cavitation can be indirectly assessed through a variety of physical and chemical methods [93]. These methods rely on detection and measurement of various features of the cavitation field. Different measurement features employed in these methods are listed in Table 2. The results of these measurements must be considered based on a basic interpretation of the cavitation process, which can be complicated. The high temperatures and pressures that

result from cavitation can lead to light emission (sonoluminescence) and increased chemical reaction rates (sonochemistry). Production of light and by-products from chemical reactions as a result of cavitation has been used to quantify cavitation activity. For example sonoluminescence and sonochemistry from an electrohydraulic shock-wave lithotripter induced cavitation field were studied as a function of spark discharge voltage (13–21 kV) and pulse-repetition frequency (0.5–2.0 Hz) [94]. It has been reported that the immediate signal was emitted within the 4 MPa peak negative pressure contour [95]. The mechanical force exerted on a steel ball by both the incident shock wave and the cavitation phenomenon have been measured using an electromagnetic probe device [96]. Detecting luminescence is possible either in an optically transparent medium or by introducing an optical probe into the system. Electron spin resonance is destructive to the sample and real-time measurement is not possible. In addition, cavitation is not directly detected in these techniques, but rather its presence is inferred from the nature of the onset of free radical production [97].

3.1.2 High-Speed Microscopy

Visualizing microbubbles is the most direct way to study their dynamics and it requires techniques of high temporal and spatial resolution. To monitor the dynamics of microbubbles activated by ultrasound at megahertz of frequency, imaging techniques with frame rates of at least several millions of frames per second are required [98]. In an *in vitro* setting, cavitation bubble behavior can be observed by means of a high-speed camera [99–101]. In order to image the bubble during the growth phase, imaging is required at length scales on the order of a millimeter and time scales on the order of tens of microseconds in a transparent media. At the nadir of the collapse, the bubble radius is less than 1 μm and the time scale of the collapse is about a nanosecond. After collapse, the remaining bubble is on the order of 10 micrometers and exists for hundreds of milliseconds before it dissolves. Therefore, it is virtually impossible to capture all bubbles dynamically considering the range of temporal and spatial scales. Bubble dynamics in water were displayed in sequential photographs for the first time by Tomita and Shima [102]. Figure 3 shows the series of photographs of the cavitation process very close to a solid surface (top frame). The time interval between each of these numbered images is 2 μs and the frame width is 1.4 mm.

One of the techniques that can be used to study the dynamic interaction between microbubbles and cell membranes is high-speed photography along with fluorescence microscopy. Instruments in this imaging system can document the event from the desired frame rate of at least 60 frame per second (fps) to extremely high frame rates up to 6.2 million frames per second (Mfps) [103]. Using this

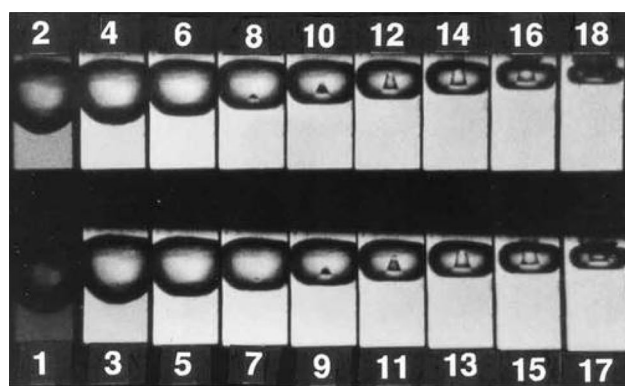


Fig. 3 A series of 18 photographs demonstrate the dynamics of the cavitation process very close to a solid surface. Reproduced from Tomita and Shima [102] with permission

technique, Ibsen et al. [104] released optical images of the interaction of a unique drug delivery vehicle (fluorescently labeled microbubbles loaded with drug) with ultrasound. Subsequently, Ibsen et al. designed an analysis system combined with a fluorescent microscope, high-speed camera, and definable pulse sequence of focused ultrasound and observed the real-time interaction between focused ultrasound, echogenic drug delivery vehicles, and live cell membranes [103]. Figure 4a shows sequential images of the interaction of focused ultrasound with a microbubble inside the echogenic drug delivery vehicle. The ruptured outer membrane of the microbubble leaves a debris field of fluorescent particles. The membrane fragmentation

generates a jet of debris in frame 3 that is followed by the diffusion of a lipid debris cloud in frame 4 [103]. A fluorescent image series of the interaction of the echogenic drug delivery vehicle with an artificial cell membrane under focused ultrasound pulsation is shown in Fig. 4b.

Another imaging technique is ultra-high-speed imaging in which the recording speed depends on the type of camera that is used. One type of that is used is Imacon 200 (DRS Hadland, Cupertino, CA, USA) that has a maximum recording speed of 200 Mfps with maximum total frame of 16. Another type of camera, the Brandaris-128 is capable of recording 128 frames with framerates up to 25 Mfps [98].

Another optical imaging technique is based on the microbubble oscillation. During an ultrasound/acoustic pulse, either exogenous or endogenous cavitation-induced microbubbles go through an expansion and contraction cycle as a result of the pressure rarefaction and compression of the applied acoustic beam. Capturing the expansion and breakup of the process requires a shutter speed on the order of tens to hundreds of nanoseconds. This can be done with a high-speed camera capable of nanosecond shutter speeds (Fig. 5a) [26]. In the absence of a nearby solid boundary, the bubble's symmetric spherical oscillation is observed using a radius-time “streak” imaging technique. In addition, the effect of ultrasound on the microbubble can be characterized by measuring the radial oscillations using a radius-time curve image of a single line through the center of the microbubble (Fig. 5b) [105].

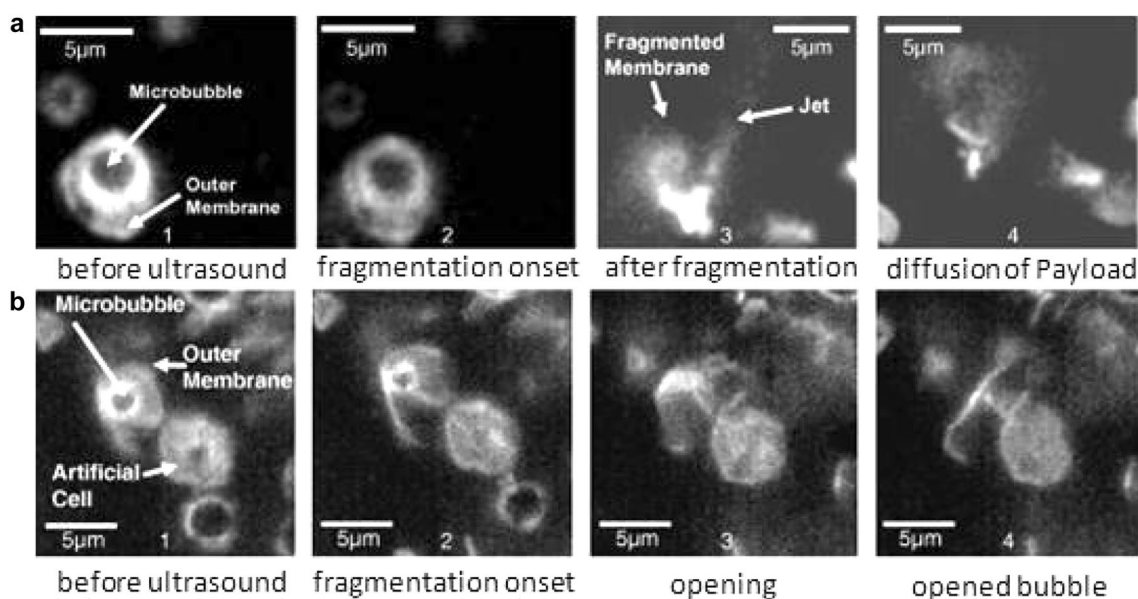


Fig. 4 **a** Sequence of fluorescent images illustrating the interaction of focused ultrasound with the microbubble inside the echogenic drug delivery vehicle and **b** sequence of fluorescent images of the interac-

tion of an echogenic drug delivery vehicle and an artificial cell membrane under ultrasound pulse sequence. Reproduced from Ibsen et al. [104] with permission

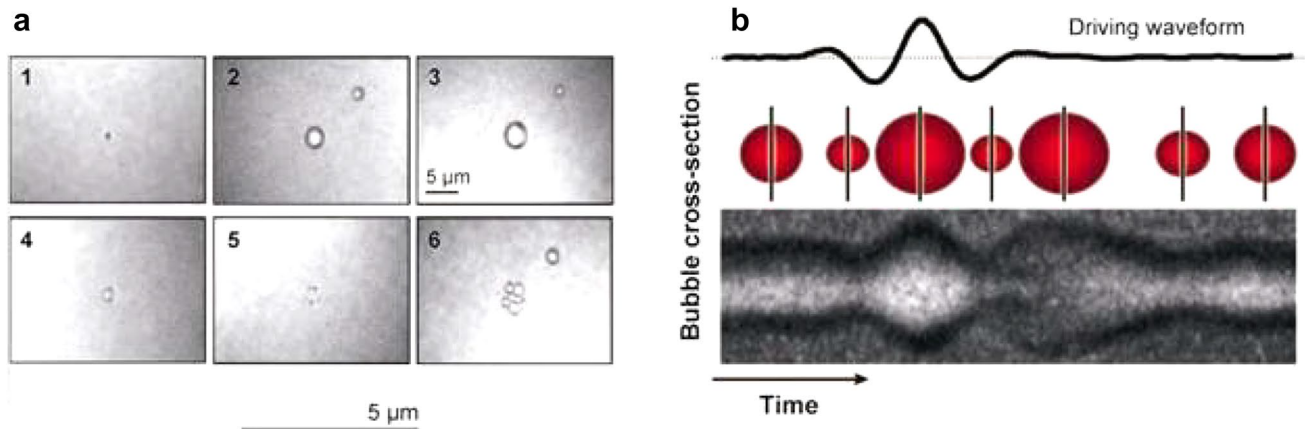


Fig. 5 **a** Sequence of two-dimensional images of a microbubble oscillation affected by a 2.25-MHz center frequency pulse ultrasound over 10 ns. **b** Radius-time image of a microbubble oscillation and a dis-

tance-time image of a single line through the center of the microbubble oscillation for characterization of the effect of ultrasound on the microbubble. Reproduced from Ferrara et al. [26] with permission

3.2 Acoustic Detection of Bubbles (Active and Passive Imaging)

An increasing number of ultrasound therapies are trying to harness the potential benefits of cavitation, both with or without intravascular ultrasound contrast agents. In order to precisely guide therapy at the focus and avoid undesirable cavitation events elsewhere within the beam path, cavitation events must be detected and mapped. Magnetic resonance imaging (MRI) can be used for anatomical targeting but is not capable of directly monitoring the effect of cavitation. In addition, cavitation does not necessarily only have a thermal effect and may happen on a very short timescale and therefore magnetic resonance thermometry is not sufficient [106]. Recently, real-time monitoring of inertial cavitation effects of microbubbles by using MRI-half-Fourier acquisition single-shot turbo spin-echo (HASTE) have been done because of water flow made by inertial cavitation generated during 2.8 MPa focused ultrasound exposure. It was reported that when microbubbles concentration increased from 0.025–0.1%, the signal intensity change of 11–55%. Their study proves the feasibility of using HASTE imaging for detection of the occurrence of inertial cavitation of microbubbles and real time signal changes monitoring during focused ultrasound [107]. Efforts are being made to detect cavitation events by ultrasound. Acoustic detection methods rely on acoustic emission created from cavitation bubbles. When a microbubble is affected by an acoustic field, two acoustic emissions are generated by the bubble: one when the bubble is hit by the acoustic wave and one when the bubble collapses. This method is powerful because it can be used in vivo (living subject). In this method, the sound waves are sent toward the cavitation field and then the sound reflections from the bubbles are picked up. Acoustic

detection is usually performed either as active cavitation detection (ACD) or passive cavitation detection (PCD) [108–110]. In ACD, one transducer sends the ultrasound wave toward the cavitation field and another transducer picks up sound reflections from cavitation bubbles, while PCD involves one or more receiving transducers for the double-bang induced acoustic emissions from cavitation bubbles [1]. In dual PCD, it is possible to use two receiving transducers and coincidentally detect a small and discrete cavitation field volume where the transducers intersect [111]. Both ACD and PCD are described in more detail elsewhere [109, 110]. In many thermal therapy applications of ultrasound where cavitation is unwanted, such a system may be used to record the broadband emissions related to inertial cavitation in order to terminate them before inadvertent tissue injury. Such a clinical ultrasound system is now commercially available (ExAblate®, Insightec, Haifa, Israel). Similar systems may be also used to monitor the broadband and harmonic emissions from the desired cavitation or intravascular ultrasound contrast agents. Real-time adjustments to the transmitted acoustic power can be performed with post-processing of signals to obtain a specific treatment effect [106]. In initial work by Hockman et al., cavitation-enhanced thermal lesioning was tested in agar gel and the process controlled using a single passive cavitation detector [112]. Passive cavitation mapping relies upon the scattering of the irradiation field by the bubble clouds, and its use to guide therapy is still in its early stages. More passive cavitation mapping systems with slightly higher coverage have been evaluated in vitro using multi elements arrays [113, 114], but for clinical use the detected area of cavitation events should be correlated with histological lesions in vivo. Currently, passive cavitation imaging is the only clinically relevant method capable of specifically monitoring cavitation events as well as providing

information about the mode and strength of the oscillations in real time [114, 115]. In this method, which mostly relies on the formation of diverging pressure waves from oscillating bubbles at frequencies different than focused ultrasound device, the final two- or three-dimensional image of the cavitation bubbles (source of emission) can be estimated through back-propagation. The application of this imaging method within a tissue phantom and an in vitro flow phantom was described in studies by Gyongy and Coussios [114] and Haworth et al. [115], respectively.

The size of the initial cavitation bubble and the sonic amplitude of the equipment that produced the cavitation bubbles affect the timing and amplitude of cavitation bubble emissions. Therefore, although acoustic detection is not able to provide information on the number and size of bubbles (this method does not image the bubbles), this method provides valuable data that can be helpful in characterizing the acoustic intensity of ultrasound therapeutic equipment (such as a lithotripter or HIFU device) in the body as well as analyzing the environment and dynamics of the cavitation field [95, 100, 111].

In 1996, Holland et al. reported the first direct evidence of in vivo cavitation from diagnostic ultrasound pulses [91]. They employed an ACD to detect short-lived cavitation in rat lung exposed to the output of a clinical diagnostic scanner (5 MHz) to identify the cause of damage. They reported observing the 30-MHz interrogating signal scattered by the short-lived bubble as a direct indication of cavitation.

There is a high demand for detection of cavitation occurrence in kidney tissue, as it is a critical step toward determining the mechanisms of tissue injury in shock wave lithotripsy. Both B-mode ultrasound [116] and focused, single-element, passive receivers [100, 117] have been studied to establish direct evidence of cavitation within the kidney. However, it is not possible when using one transducer to differentiate between the occurrence of cavitation within kidney parenchyma and that in the fluid spaces of the collecting system [111]. Bailey et al. developed a cavitation detection system in which dual passive receivers and B-mode ultrasound were coaligned with the focal point of a Dornier HM3 lithotripter to interrogate cavitation occurrence within the renal parenchyma as well as in the collecting system during lithotripsy [92]. Their study provided direct evidence of cavitation occurrence in the renal parenchyma. Monitoring of thermal lesioning with B-mode ultrasound requires interleaving with the treatment beam. It typically displays a hyperechoic area secondary to bubble formation, but these bubbles may be induced through boiling of the tissues and are often undesirable because scattering from the larger bubbles can extend the area of coagulation into the prefocal zone [118].

Ultrasound-guided localized detection of cavitation during lithotripsy has been also studied in vivo by a dual passive

cavitation detection (DPCD) system in pig kidney [119]. The cavitation appeared as a hyperecho on the B-mode image of the kidney. Following imaging, dissection of the kidney revealed a V-shaped lesion created by the transducer.

Measurement techniques currently used to measure key acoustic output and exposure parameters such as cavitation that might be relevant to biological acoustic effects are performed in water. The acoustic properties of the water medium are very different from those of tissue [120]. Therefore, estimation of in situ ultrasound exposure at a site of interest within the patient is very difficult [121].

3.3 Scattering Methods

3.3.1 Laser Scattering Technique

Another method of measuring cavitation is laser scattering of single bubbles. In this technique, a laser beam illuminates the bubble, and a photodetector collects the scattered light from the bubble. For a single spherical-shaped bubble, the amplitude of the scattered light changes predictably with the radius of the bubble. In this method, most of the temporal and spatial scales related to the dynamics of a cavitation bubble can be captured.

3.3.2 Synchrotron X-Ray Imaging Technique

Another recent attempt to study and directly visualize microbubbles or ultrasound-induced cavitation bubbles uses synchrotron X-ray imaging. Among the different synchrotron X-ray imaging techniques, X-ray phase contrast imaging (PCI) and analyzer based imaging (ABI) have shown potential for detection of microbubbles. In ABI, which is a phase-sensitive imaging technique, a very bright and highly collimated X-ray beam is sent toward the cavitation field where small structures such as microbubbles will refract and scatter the X-rays through small angles. Using ABI, these small angles can create contrast based on the very narrow reflectivity curve of the analyzer crystal. The refracted and scattered X-ray beams are collected at the detector behind the sample. This technique is also capable of in vivo detection of microbubbles deep in tissue. The advantage of using an X-ray technique such as ABI is that the microbubble properties can be visualized in vivo without any effect of the X-rays on the bubbles or vice versa. In 2010, Arfelli et al. studied the feasibility of visualizing the microbubbles as contrast agents using ABI [122]. These microbubbles are invisible in conventional X-ray absorption techniques. [122]. Arfelli and colleagues evaluated the possibility of visualizing two different microbubble contrast agents commonly used in clinical ultrasound examination (Levovist® and Optison™) in sizes ranges between 1 and 8 μm in custom-made phantoms using ABI [122]. They reported that ABI demonstrated high

visibility of the details with stronger contrast than normal X-ray absorption techniques. On the other hand, because microbubbles contrast agents can act as an X-ray lens, with individual microbubbles refracting and a population of them scattering the X-rays, PCI was considered a potential synchrotron based X-ray imaging technique for detection of microbubbles by Millard et al. [123]. All of these studies focused on visualization and monitoring exogenous microbubbles contrast agents that usually contain either air or perfluorocarbon gas with premade shells and have prolonged longevity because of their low solubility (they are stable up to several minutes). Recently, another study attempted to use an ABI modality to visualize microbubbles/ultrasound-induced cavitation bubbles (endogenous microbubbles) at the biomedical imaging and therapy (BMIT) beamline at the Canadian Light Source (CLS). Direct visualization of ultrasound-induced cavitation bubbles was achieved at 20 kHz and 130 W [124] and at a therapeutic clinical system level of 0.88 MHz and 14 W [125]. Figure 6 shows the spatial structural pattern of ultrasound-induced cavitation bubbles from a therapeutic clinical system using the ABI technique. The cavitation bubbles appear in a periodic pattern and the distance between two consecutive parallel lines of the pattern (Fig. 6b) was measured as 0.99 mm. This work demonstrated that ABI has significant utility with respect to detecting and visualizing microbubbles/cavitation bubbles.

Table 1 gives a summary of different cavitation bubble detection techniques and their related properties. A comparison of different imaging methods for visualizing microbubbles in terms of practical advantages and disadvantages is given in Table 2.

4 Summary and Recommendations for Future Research

Advances in diagnostic ultrasound imaging have expanded its use to obstetrics and fetal, embryo, or follicle development imaging. In recent decades, large numbers of fetuses have been scanned on a routine basis. Furthermore, the

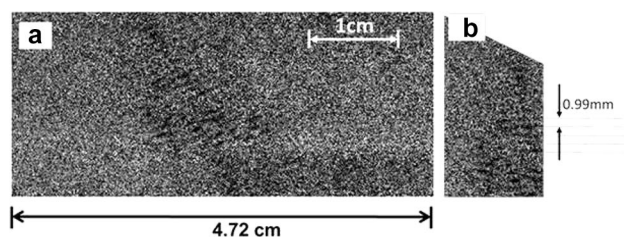


Fig. 6 **a** ABI of the whole ultrasound beam of a therapeutic ultrasound system at 0.8835 MHz and 14 W and **b** the sequence and location of cavitation bubbles, with approximately the same interval between sequences. Reproduced from Izadifar et al. [125]

ability of ultrasound to penetrate deep into soft tissue at low frequencies (0.7–3.3 MHz) has prompted its use for non-invasive therapies such as hyperthermia and extracorporeal shock wave lithotripsy. Although the therapeutic and imaging mechanisms of ultrasound are based on the interaction of sound waves with tissue, cavitation can also occur during ultrasound imaging and therapy and result in hazardous bioeffects on tissues. It is essential to conduct fundamental cavitation studies to identify the threshold parameters of bubble formation and collapse that give rise to bioeffects. These findings may improve standards of ultrasound safety for a range of therapeutic and diagnostic applications, such as fetal and embryo imaging. High levels of ultrasound exposure have well-recognized acute harmful effects (measures and guidelines exist for medical uses to avoid these). At lower levels of exposure, especially diagnostic ultrasound levels, there is no established evidence of specific hazards; however, few data are available to confirm this, especially over the long term. This concern has been also stated in the report of the advisory group on non-ionizing radiation [121]. In addition, microbubbles and cavitation are playing increasingly important roles in clinical applications of both diagnostic and therapeutic ultrasound. Microbubbles have been used as contrast agents in clinics for decades. They have also been developed as vehicles loaded with therapeutic agents for drug delivery and gene therapy. These vehicles are traced to the target site using diagnostic ultrasound [126]. The use of microbubbles in the future for drug/gene delivery is promising. Also, controlled cavitation is being studied as a means to enhance the speed and efficacy of treatment in HIFU surgery and lithotripsy.

Monitoring the cavitation/microbubbles in different tissues in vivo for different treatment conditions will give more insight into the conditions under which ultrasound can be used safely. Furthermore, for high power devices used in therapeutic and surgical applications in which the purpose is to deliver high-intensity ultrasound to a target tissue, the concerns apart from safety are mainly with respect to accurate targeting of the desired target volume to avoid damaging other tissues. Detecting the existence and location of microbubbles/cavitation bubbles in tissue will provide more insight for more precise application of therapeutic ultrasounds such as HIFU, lithotripsy, and drug/gene delivery. Various imaging techniques have been evaluated for detection of microbubbles/cavitation bubbles. Each imaging technique has advantages and disadvantages for studying microbubble behavior. However, their application in vivo and particularly deep in tissue for locating microbubbles in the human body is an aspect of interest for future clinical study. Among the different microbubble detection techniques, acoustic and X-ray detection techniques have advantages for use in vivo and particularly in the human body. Also, X-ray imaging techniques can detect microbubbles without

Table 1 Summary of different cavitation bubble detection techniques and related properties

Detection method	Measurement modality	Measured features	Compatible media	Advantages	Limitations
Optical	Physical/chemical response detection	Secondary	Transparent [97]	Providing quantitative measurements	Complication in interpretation of results [1] Limited application to in vivo systems [97] Potential destructive effect (i.e., ESR) [97] Do not allow real-time measurement [97]
	High speed microscopy imaging	Primary-direct imaging	Transparent [102–104, 132]	Reveals dynamics of the cavitation process Visualization of cavitation bubble-adjacent surface interaction at sub-microsecond resolution [98, 132]	Not applicable to in vivo system Limited practicality Limited depth of field [1] Limited field of view; unable to image bubble clouds [1]
Acoustic detection	Active and Passive imaging	Secondary	Opaque [132, 115] Transparent [112, 126]	2D [120] and 3D imaging of cavitation field Applicable to clinical devices to study cavitation/microbubble contrast agent mechanism [133]	Possible interference with cavitation process Difficult estimation of in situ ultrasound exposure within the patient [121]
	Scattering-based detection	Primary	Transparent [1]	Very precise measurement of spherical-shaped bubble dynamics [94]	Limited to small sample volume Requires unrestricted visual access at high magnification Unable to give qualitative information about bubble clouds or non-spherical bubbles [1]
Synchrotron X-ray imaging	Laser scattering	Primary	Opaque [122] Transparent [124, 125]	Does not interfere with cavitation process Provides qualitative information about the microbubble dynamics (location and their existence) Enables qualitative monitoring of microbubbles [123]	Unable to provide information on the dynamics, size, or behavior of a single bubble [124, 125]

Table 2 Advantages and disadvantages of different imaging modalities for visualizing microbubbles

Imaging modalities	In vivo	In vitro	Dynamic detection	Passive detection	Sample volume independent	Sample area independent	Qualitative assessment	Presence of ionizing radiation
High-speed photography	—	+	+	—	—	—	—	—
Laser scattering	—	+	+	—	—	+	—	—
Chemical and physical methods	—	+	—	—	+	+	—	+
Acoustic detection	+	+	—	—	+	+	+	—
Synchrotron X-ray detection	+	+	—	+	+	+	+	+

— indicates the absence and + indicates the presence of each listed assessment capability in detection modalities

disturbing them, which is not the case with acoustic detection methods such as ACD and PCD.

The ABI technique has shown promise for direct visualization of cavitation bubbles deep in tissue through the use of a very bright and collimated X-ray beam from a synchrotron. This imaging technique is under study by the authors at the BMIT beamline at the CLS [124]. ABI has a significant and demonstrated ability to image structures that feature interfaces between materials of different density, such as that between air and water in a bubble. The advantage of using an X-ray technique is the ability to visualize the cavitation bubbles/microbubbles in an optically opaque material, such as tissue, without any physical interaction with the bubbles.

There is no doubt that continuous discoveries and developments in microbubble detection modalities will lead to safer and more efficient therapeutic and diagnostic equipment and also more precise microbubble applications in medicine.

Acknowledgements The authors acknowledge the University of Saskatchewan Dean's scholarship program (ZI), the Natural Science and Engineering Research Council of Canada (NSERC) Discovery Grant program, and the Canada Research Chairs program (DC).

Author Contributions ZI wrote and organized the manuscript, PB and DC read and revised the manuscript.

Funding This work was supported in part by a NSERC Discovery Grant, the Canada Research Chairs Program, and the University of Saskatchewan.

Compliance with Ethical Standards

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this paper.

References

- Cleveland, R. O., & McAteer, J. A. (2012). Extracorporeal shock wave lithotripsy: the physics of shock wave lithotripsy. In B. C. Decker (Ed.), *Smith's Textbook on Endourology*. PMPH: Ontario.
- Chatel, G., & Colmenares, J. C. (2017). Sonochemistry: from basic principles to innovative applications. *Topics in Current Chemistry*. <https://doi.org/10.1007/s41061-016-0096-1>.
- Holland, C. K., Deng, C. X., Apfel, R. E., Alderman, J. L., Fernandez, L. A., & Taylor, K. J. (1996). Direct evidence of cavitation in vivo from diagnostic ultrasound. *Ultrasound in Medicine and Biology*, 22(7), 917–925.
- Miller, D. L., Smith, N. B., Bailey, M. R., Czarnota, G. J., Hynnen, K., & Makin, I. R. (2012). Overview of therapeutic ultrasound applications and safety considerations. *Journal of Ultrasound in Medicine*, 31(4), 623–634.
- Atchley, A. A., & Prosperetti, A. (1989). The crevice model of bubble nucleation. *The Journal of the Acoustical Society of America*, 86(3), 1065–1084. <https://doi.org/10.1121/1.398098>.
- Fuchs, J. (2011) Ultrasonics—Number and Size of Cavitation Bubbles. <http://www.ctgclean.com/tech-blog/2011/12/ultrasonics-number-and-size-of-cavitation-bubbles/>. Accessed 28 Sept 2015.
- Leong, T., Ashokkumar, M., & Kentish, S. (2011). The fundamentals of power ultrasound—a review. *Acoustics Australia*, 39(2), 54–63.
- Church, C. C., & Carstensen, E. L. (2001). “Stable” inertial cavitation. *Ultrasound in Medicine and Biology*, 27(10), 1435–1437.
- Luque-García, J. L., & de Castro, M. D. L. (2003). Ultrasound: a powerful tool for leaching. *TrAC Trends in Analytical Chemistry*, 22(1), 41–47. [https://doi.org/10.1016/S0165-9936\(03\)00102-X](https://doi.org/10.1016/S0165-9936(03)00102-X).
- Suslick, K. (1994). *The yearbook of science and the future* (p. 138). Chicago: Encyclopedia Britannica.
- Noltingk, B. E., & Neppiras, E. A. (1950). Cavitation produced by ultrasonics. *Proceedings of the Physical Society Section B*, 63(9), 674.
- Plesset, M. (1949). The dynamics of cavitation bubbles. *Journal of Applied Mechanics*, 16, 277–282.
- Apfel, R. E., & Holland, C. K. (1991). Gauging the likelihood of cavitation from short-pulse, low-duty cycle diagnostic ultrasound. *Ultrasound in Medicine and Biology*, 17(2), 179–185.
- Holt, R. G., & Roy, R. A. (2001). Measurements of bubble-enhanced heating from focused, MHz-frequency ultrasound in a tissue-mimicking material. *Ultrasound in Medicine and Biology*, 27(10), 1399–1412.
- Nyborg, W. L. (1998). *Acoustic streaming nonlinear acoustics*. San Diego: Academic Press.
- Dijkmans, P. A., Juffermans, L. J., Musters, R. J., van Wamel, A., ten Cate, F. J., van Gilst, W., et al. (2004). Microbubbles and ultrasound: from diagnosis to therapy. *Eur J Echocardiogr.*, 5(4), 245–256. <https://doi.org/10.1016/j.euje.2004.02.001>.
- Skyba, D. M., Price, R. J., Linka, A. Z., Skalak, T. C., & Kaul, S. (1998). Direct in vivo visualization of intravascular

- destruction of microbubbles by ultrasound and its local effects on tissue. *Circulation*, 98(4), 290–293.
18. Nejad, S. M., Hosseini, H., Akiyama, H., & Tachibana, K. (2016). Reparable cell sonoporation in suspension: theranostic potential of microbubble. *Theranostics*, 6(4), 446.
19. Fan, Z., Chen, D., & Deng, C. (2013). Improving ultrasound gene transfection efficiency by controlling ultrasound excitation of microbubbles. *Journal of Controlled Release*, 170(3), 401–413.
20. Helfield, B., Chen, X., Watkins, S. C., & Villanueva, F. S. (2016). Biophysical insight into mechanisms of sonoporation. *Proceedings of the National Academy of Sciences*, 113(36), 9983–9988.
21. Matlaga, B. R., McAteer, J. A., Connors, B. A., Handa, R. K., Evan, A. P., Williams, J. C., et al. (2008). Potential for cavitation-mediated tissue damage in shockwave lithotripsy. *Journal of Endourology*, 22(1), 121–126.
22. Lawrie, A., Briskin, A. F., Francis, S. E., Tayler, D. I., Chamberlain, J., Crossman, D. C., et al. (1999). Ultrasound enhances reporter gene expression after transfection of vascular cells in vitro. *Circulation*, 99(20), 2617–2620.
23. Stride, E., & Saffari, N. (2003). On the destruction of microbubble ultrasound contrast agents. *Ultrasound in Medicine and Biology*, 29(4), 563–573. [https://doi.org/10.1016/s0301-5629\(02\)00787-1](https://doi.org/10.1016/s0301-5629(02)00787-1).
24. Hernot, S., & Klivanov, A. L. (2008). Microbubbles in ultrasound-triggered drug and gene delivery. *Advanced Drug Delivery Reviews*, 60(10), 1153–1166. <https://doi.org/10.1016/j.addr.2008.03.005>.
25. Basta, G., Venneri, L., Lazzerini, G., Pisanis, E., Pianelli, M., Vesentini, N., et al. (2003). In vitro modulation of intracellular oxidative stress of endothelial cells by diagnostic cardiac ultrasound. *Cardiovascular Research*, 58(1), 156–161.
26. Ferrara, K., Pollard, R., & Borden, M. (2007). Ultrasound microbubble contrast agents: fundamentals and application to gene and drug delivery. *Annual Review of Biomedical Engineering*, 9, 415–447. <https://doi.org/10.1146/annurev.bioeng.8.061505.095852>.
27. Nanda, N., Schlieff, R., & Goldberg, B. B. (1997). *Advances in echo imaging using contrast enhancement* (2nd ed.). Dordrecht: Kluwer Academic.
28. Mayer, S., & Grayburn, P. A. (2001). Myocardial contrast agents: recent advances and future directions. *Progress in Cardiovascular Diseases*, 44(1), 33–44. <https://doi.org/10.1053/pcad.2001.26438>.
29. Bouakaz, A., de Jong, N., Cachard, C., & Jouini, K. (1998). On the effect of lung filtering and cardiac pressure on the standard properties of ultrasound contrast agent. *Ultrasonics*, 36(1–5), 703–708.
30. Unger, E. C., Hersh, E., Vannan, M., Matsunaga, T. O., & McCreery, M. (2001). Local drug and gene delivery through microbubbles. *Progress in Cardiovascular Diseases*, 44(1), 45–54. <https://doi.org/10.1053/pcad.2001.26443>.
31. Mulvagh, S. L., DeMaria, A. N., Feinstein, S. B., Burns, P. N., Kaul, S., Miller, J. G., et al. (2000). Contrast echocardiography: Current and future applications. *Journal of the American Society of Echocardiography*, 13(4), 331–342. <https://doi.org/10.1067/mje.2000.105462>.
32. Liang, H., Tang, J., & Halliwell, M. (2010). Sonoporation, drug delivery, and gene therapy. *Proceedings of the Institution of Mechanical Engineers, Part H, Journal of Engineering in Medicine*, 224(2), 343–361.
33. Collis, J., Manasseh, R., Liovic, P., Tho, P., Ooi, A., Petkovic-Duran, K., et al. (2010). Cavitation microstreaming and stress fields created by microbubbles. *Ultrasonics*, 50(2), 273–279.
34. Wu, J., Ross, J. P., & Chiu, J.-F. (2002). Reparable sonoporation generated by microstreaming. *The Journal of the Acoustical Society of America*, 111(3), 1460–1464.
35. Liu, H.-L., Fan, C.-H., Ting, C.-Y., & Yeh, C.-K. (2014). Combining microbubbles and ultrasound for drug delivery to brain tumors: current progress and overview. *Theranostics*, 4(4), 432.
36. Chu, P. C., Liu, H. L., Lai, H. Y., Lin, C. Y., Tsai, H. C., & Pei, Y. C. (2015). Neuromodulation accompanying focused ultrasound-induced blood-brain barrier opening. *Scientific Reports*, 5, 15477. <https://doi.org/10.1038/srep15477>.
37. Yoo, S.-S., Bystritsky, A., Lee, J.-H., Zhang, Y., Fischer, K., Min, B.-K., et al. (2011). Focused ultrasound modulates region-specific brain activity. *Neuroimage*, 56(3), 1267–1275.
38. Bystritsky, A., Korb, A. S., Douglas, P. K., Cohen, M. S., Melega, W. P., Mulgaonkar, A. P., et al. (2011). A review of low-intensity focused ultrasound pulsation. *Brain Stimulation*, 4(3), 125–136. <https://doi.org/10.1016/j.brs.2011.03.007>.
39. Rezayat, E., & Toostani, I. G. (2016). A review on brain stimulation using low intensity focused ultrasound. *Basic and Clinical Neuroscience*, 7(3), 187.
40. Hynnen, K., McDannold, N., Vykhodtseva, N., & Jolesz, F. A. (2001). Noninvasive MR imaging guided focal opening of the blood-brain barrier in rabbits I. *Radiology*, 220(3), 640–646.
41. Chu, P. C., Chai, W. Y., Tsai, C. H., Kang, S. T., Yeh, C. K., & Liu, H. L. (2016). Focused ultrasound-induced blood-brain barrier opening: association with mechanical index and cavitation index analyzed by dynamic contrast-enhanced magnetic-resonance imaging. *Scientific Reports*, 6, 33264. <https://doi.org/10.1038/srep33264>.
42. Porter, T. R., Hiser, W. L., Kricsfeld, D., Deligonul, U., Xie, F., Iversen, P., et al. (2001). Inhibition of carotid artery neointimal formation with intravenous microbubbles. *Ultrasound in Medicine and Biology*, 27(2), 259–265.
43. Price, R. J., & Kaul, S. (2002). Contrast ultrasound targeted drug and gene delivery: an update on a new therapeutic modality. *J Cardiovascular Pharmacology Therapy*, 7(3), 171–180.
44. Ng, K. Y., & Liu, Y. (2002). Therapeutic ultrasound: its application in drug delivery. *Medicinal Research Reviews*, 22(2), 204–223. <https://doi.org/10.1002/med.10004>.
45. Lindner, J. R. (2002). Evolving applications for contrast ultrasound. *American Journal of Cardiology*, 90(10A), 72J–80J.
46. Unger, E. C., Matsunaga, T. O., McCreery, T., Schumann, P., Sweitzer, R., & Quigley, R. (2002). Therapeutic applications of microbubbles. *European Journal of Radiology*, 42(2), 160–168. [https://doi.org/10.1016/s0720-048x\(01\)00455-7](https://doi.org/10.1016/s0720-048x(01)00455-7).
47. Tachibana, K., & Tachibana, S. (2001). The use of ultrasound for drug delivery. *Echocardiography*, 18(4), 323–328. <https://doi.org/10.1046/j.1540-8175.2001.00323.x>.
48. Lindner, J. R., & Kaul, S. (2001). Delivery of drugs with ultrasound. *Echocardiography*, 18(4), 329–337. <https://doi.org/10.1046/j.1540-8175.2001.00329.x>.
49. McDannold, N., Arvanitis, C. D., Vykhodtseva, N., & Livingstone, M. S. (2012). Temporary disruption of the blood–brain barrier by use of ultrasound and microbubbles: safety and efficacy evaluation in rhesus macaques. *Cancer Research*, 72(14), 3652–3663.
50. Hynnen, K., McDannold, N., Vykhodtseva, N., Raymond, S., Weissleder, R., Jolesz, F. A., et al. (2006). Focal disruption of the blood–brain barrier due to 260-kHz ultrasound bursts: a method for molecular imaging and targeted drug delivery. *Journal of Neurosurgery*, 105(3), 445–454.
51. Tartis, M. S., McCallan, J., Lum, A. F., LaBell, R., Stieger, S. M., Matsunaga, T. O., et al. (2006). Therapeutic effects of paclitaxel-containing ultrasound contrast agents. *Ultrasound in Medicine and Biology*, 32(11), 1771–1780. <https://doi.org/10.1016/j.ultrasmedbio.2006.03.017>.
52. Chen, S., J-h, Ding, Bekerajian, R., B-z, Yang, Shohet, R. V., Johnston, S. A., et al. (2006). Efficient gene delivery to pancreatic islets with ultrasonic microbubble destruction technology.

- Proceedings of the National Academy of Sciences*, 103(22), 8469–8474.
53. Hauff, P., Seemann, S., Reszka, R., Schultze-Mosgau, M., Reinhardt, M., Buzasi, T., et al. (2005). Evaluation of gas-filled microparticles and sonoporation as gene delivery system: feasibility study in rodent tumor models I. *Radiology*, 236(2), 572–578.
 54. Price, R. J., Skyba, D. M., Kaul, S., & Skalak, T. C. (1998). Delivery of colloidal particles and red blood cells to tissue through microvessel ruptures created by targeted microbubble destruction with ultrasound. *Circulation*, 98(13), 1264–1267.
 55. Miller, D. L., & Quddus, J. (2000). Diagnostic ultrasound activation of contrast agent gas bodies induces capillary rupture in mice. *Proceedings of the National Academy of Sciences*, 97(18), 10179–10184.
 56. Dimcevski, G., Kotopoulis, S., Bjånes, T., Hoem, D., Schjøtt, J., Gjertsen, B. T., et al. (2016). A human clinical trial using ultrasound and microbubbles to enhance gemcitabine treatment of inoperable pancreatic cancer. *Journal of Controlled Release*, 243, 172–181.
 57. Kotopoulis, S., Dimcevski, G., Gilja, O. H., Hoem, D., & Postema, M. (2013). Treatment of human pancreatic cancer using combined ultrasound, microbubbles, and gemcitabine: a clinical case study. *Medical Physics*, 40(7), 072902.
 58. Wu, J. R. (1998). Temperature rise generated by ultrasound in the presence of contrast agent. *Ultrasound in Medicine and Biology*, 24(2), 267–274. [https://doi.org/10.1016/s0301-5629\(97\)00246-9](https://doi.org/10.1016/s0301-5629(97)00246-9).
 59. Miller, D. L., & Gies, R. A. (1998). Enhancement of ultrasonically-induced hemolysis by perfluorocarbon-based compared to air-based echo-contrast agents. *Ultrasound in Medicine and Biology*, 24(2), 285–292. [https://doi.org/10.1016/s0301-5629\(97\)00267-6](https://doi.org/10.1016/s0301-5629(97)00267-6).
 60. Poliachik, S. L., Chandler, W. L., Mourad, P. D., Bailey, M. R., Bloch, S., Cleveland, R. O., et al. (1999). Effect of high-intensity focused ultrasound on whole blood with and without microbubble contrast agent. *Ultrasound in Medicine and Biology*, 25(6), 991–998. [https://doi.org/10.1016/s0301-5629\(99\)00043-5](https://doi.org/10.1016/s0301-5629(99)00043-5).
 61. Bekereditian, R., Chen, S., Frenkel, P. A., Grayburn, P. A., & Shohet, R. V. (2003). Ultrasound-targeted microbubble destruction can repeatedly direct highly specific plasmid expression to the heart. *Circulation*, 108(8), 1022–1026. <https://doi.org/10.1161/01.cir.0000084535.35435.ae>.
 62. Frenkel, P. A., Chen, S., Thai, T., Shohet, R. V., & Grayburn, P. A. (2002). DNA-loaded albumin microbubbles enhance ultrasound-mediated transfection in vitro. *Ultrasound in Medicine and Biology*, 28(6), 817–822.
 63. Shohet, R. V., Chen, S., Zhou, Y. T., Wang, Z., Meidell, R. S., Unger, R. H., et al. (2000). Echocardiographic destruction of albumin microbubbles directs gene delivery to the myocardium. *Circulation*, 101(22), 2554–2556.
 64. Smeenge, M., Tranquart, F., Mannaerts, C. K., de Reijke, T. M., van de Vijver, M. J., Laguna, M. P., et al. (2017). first-in-human ultrasound molecular imaging with a VEGFR2-specific ultrasound molecular contrast agent (BR55) in prostate cancer: a safety and feasibility pilot study. *Investigative Radiology*, 52, 419.
 65. Willmann, J. K., Bonomo, L., Carla Testa, A., Rinaldi, P., Rindi, G., Valluru, K. S., et al. (2017). Ultrasound molecular imaging with BR55 in patients with breast and ovarian lesions: first-in-human results. *Journal of Clinical Oncology*, 35, 2133.
 66. Villanueva, F. S., Jankowski, R. J., Klibanov, S., Pina, M. L., Alber, S. M., Watkins, S. C., et al. (1998). Microbubbles targeted to intercellular adhesion molecule-1 bind to activated coronary artery endothelial cells. *Circulation*, 98(1), 1–5.
 67. Weller, G. E., Lu, E., Csikari, M. M., Klibanov, A. L., Fischer, D., Wagner, W. R., et al. (2003). Ultrasound imaging of acute cardiac transplant rejection with microbubbles targeted to intercellular adhesion molecule-1. *Circulation*, 108(2), 218–224. <https://doi.org/10.1161/01.cir.0000080287.74762.60>.
 68. Ellegala, D. B., Leong-Poi, H., Carpenter, J. E., Klibanov, A. L., Kaul, S., Shaffrey, M. E., et al. (2003). Imaging tumor angiogenesis with contrast ultrasound and microbubbles targeted to alpha(v)beta3. *Circulation*, 108(3), 336–341. <https://doi.org/10.1161/01.cir.0000080326.15367.0c>.
 69. Lindner, J. R., Coggins, M. P., Kaul, S., Klibanov, A. L., Brandenburger, G. H., & Ley, K. (2000). Microbubble persistence in the microcirculation during ischemia/reperfusion and inflammation is caused by integrin- and complement-mediated adherence to activated leukocytes. *Circulation*, 101(6), 668–675.
 70. Lindner, J. R., Dayton, P. A., Coggins, M. P., Ley, K., Song, J., Ferrara, K., et al. (2000). Noninvasive imaging of inflammation by ultrasound detection of phagocytosed microbubbles. *Circulation*, 102(5), 531–538.
 71. Christiansen, J. P., Leong-Poi, H., Klibanov, A. L., Kaul, S., & Lindner, J. R. (2002). Noninvasive imaging of myocardial reperfusion injury using leukocyte-targeted contrast echocardiography. *Circulation*, 105(15), 1764–1767.
 72. Schumann, P. A., Christiansen, J. P., Quigley, R. M., McCreary, T. P., Sweitzer, R. H., Unger, E. C., et al. (2002). Targeted-microbubble binding selectively to GPIIb/IIIa receptors of platelet thrombi. *Investigative Radiology*, 37(11), 587–593. <https://doi.org/10.1097/01.rli.0000031077.17751.b2>.
 73. Akiyama, M., Ishibashi, T., Yamada, T., & Furuhashi, H. (1998). Low-frequency ultrasound penetrates the cranium and enhances thrombolysis in vitro. *Neurosurgery*, 43(4), 828–832. <https://doi.org/10.1097/00006123-199810000-00062>.
 74. Harpaz, D., Chen, X. C., Francis, C. W., Marder, V. J., & Meltzer, R. S. (1993). Ultrasound enhancement of thrombolysis and reperfusion in vitro. *Journal of the American College of Cardiology*, 21(6), 1507–1511.
 75. Kornowski, R., Meltzer, R. S., Chernine, A., Vered, Z., & Battler, A. (1994). Does external ultrasound accelerate thrombolysis? Results from a rabbit model. *Circulation*, 89(1), 339–344.
 76. Larsson, J., Carlson, J., & Olsson, S. B. (1998). Ultrasound enhanced thrombolysis in experimental retinal vein occlusion in the rabbit. *British Journal of Ophthalmology*, 82(12), 1438–1440. <https://doi.org/10.1136/bjo.82.12.1438>.
 77. Lauer, C. G., Burge, R., Tang, D. B., Bass, B. G., Gomez, E. R., & Alving, B. M. (1992). Effect of ultrasound on tissue-type plasminogen activator-induced thrombolysis. *Circulation*, 86(4), 1257–1264.
 78. Luo, H., Nishioka, T., Fishbein, M. C., Cercek, B., Forrester, J. S., Kim, C. J., et al. (1996). Transcutaneous ultrasound augments lysis of arterial thrombi in vivo. *Circulation*, 94(4), 775–778.
 79. Riggs, P. N., Francis, C. W., Bartos, S. R., & Penney, D. P. (1997). Ultrasound enhancement of rabbit femoral artery thrombolysis. *Cardiovascular Surgery*, 5(2), 201–207.
 80. Rassin, T., Desmet, W., Piessens, J., & Rosenschein, U. (2000). Ultrasound thrombolysis in stent thrombosis. *Catheterization and Cardiovascular Interventions*, 51(3), 332–334.
 81. Porter, T. R., & Xie, F. (2001). Ultrasound, microbubbles, and thrombolysis. *Progress in Cardiovascular Diseases*, 44(2), 101–110. <https://doi.org/10.1053/pcad.2001.26441>.
 82. Tachibana, K., & Tachibana, S. (1995). Albumin microbubble echo-contrast material as an enhancer for ultrasound accelerated thrombolysis. *Circulation*, 92(5), 1148–1150.
 83. Kondo, I., Mizushige, K., Ueda, T., Masugata, H., Ohmori, K., & Matsuo, H. (1999). Histological observations and the process of ultrasound contrast agent enhancement of tissue plasminogen activator thrombolysis with ultrasound exposure. *Japanese Circulation Journal*, 63(6), 478–484.

84. Nishioka, T., Luo, H., Fishbein, M. C., Cercek, B., Forrester, J. S., Kim, C. J., et al. (1997). Dissolution of thrombotic arterial occlusion by high intensity, low frequency ultrasound and dodecafluoropentane emulsion: An in vitro and in vivo study. *Journal of the American College of Cardiology*, 30(2), 561–568. [https://doi.org/10.1016/s0735-1097\(97\)00182-4](https://doi.org/10.1016/s0735-1097(97)00182-4).
85. Porter, T. R., LeVeen, R. F., Fox, R., Kricsfeld, A., & Xie, F. (1996). Thrombolytic enhancement with perfluorocarbon-exposed sonicated dextrose albumin microbubbles. *American Heart Journal*, 132(5), 964–968. [https://doi.org/10.1016/s0002-8703\(96\)90006-x](https://doi.org/10.1016/s0002-8703(96)90006-x).
86. Mizushige, K., Kondo, I., Ohmori, K., Hirao, K., & Matsuo, H. (1999). Enhancement of ultrasound-accelerated thrombolysis by echo contrast agents: dependence on microbubble structure. *Ultrasound in Medicine and Biology*, 25(9), 1431–1437. [https://doi.org/10.1016/s0301-5629\(99\)00095-2](https://doi.org/10.1016/s0301-5629(99)00095-2).
87. Nacu, A., Kvistad, C. E., Naess, H., Øygarden, H., Logallo, N., Assmus, J., et al. (2016). NOR-SASS (Norwegian Sonothrombolysis in Acute Stroke Study). *Stroke*, 48, 335.
88. Kobayashi, N., Yasu, T., Yamada, S., Kudo, N., Kuroki, M., Kawakami, M., et al. (2002). Endothelial cell injury in venule and capillary induced by contrast ultrasonography. *Ultrasound in Medicine and Biology*, 28(7), 949–956. [https://doi.org/10.1016/s0301-5629\(02\)00532-x](https://doi.org/10.1016/s0301-5629(02)00532-x).
89. Cootney, R. W. (2001). Ultrasound imaging: principles and applications in rodent research. *ILAR Journal*, 42(3), 233–247.
90. O'Reilly, M. A., Hough, O., & Hynynen, K. (2017). Blood-brain barrier closure time after controlled ultrasound-induced opening is independent of opening volume. *Journal of Ultrasound in Medicine*, 36(3), 475–483.
91. Holland, C. K., Deng, C. X., Apfel, R. E., Alderman, J. L., Fernandez, L. A., & Taylor, K. J. (1996). Direct evidence of cavitation in vivo from diagnostic ultrasound. *Ultrasound in Medicine and Biology*, 22(7), 917–925.
92. Bailey, M. R., Pishchalnikov, Y. A., Sapozhnikov, O. A., Cleveland, R. O., McAteer, J. A., Miller, N. A., et al. (2005). Cavitation detection during shock-wave lithotripsy. *Ultrasound in Medicine and Biology*, 31(9), 1245–1256. <https://doi.org/10.1016/j.ultrasmedbio.2005.02.017>.
93. Leighton, T. G. (1994). *The acoustic bubble*. San Diego: Academic Press.
94. Matula, T. J., Hilmo, P. R., Bailey, M. R., & Crum, L. A. (2002). In vitro sonoluminescence and sonochemistry studies with an electrohydraulic shock-wave lithotripter. *Ultrasound in Medicine and Biology*, 28(9), 1199–1207.
95. Coleman, A. J., Whitlock, M., Leighton, T., & Saunders, J. E. (1993). The spatial distribution of cavitation induced acoustic emission, sonoluminescence and cell lysis in the field of a shock wave lithotripter. *Physics in Medicine and Biology*, 38(11), 1545–1560.
96. Pye, S. D., & Dineley, J. A. (1999). Characterization of cavitation activity in lithotripsy fields using a robust electromagnetic probe. *Ultrasound in Medicine and Biology*, 25(3), 451–471.
97. Atchley, A., Frizzell, L., Apfel, R., Holland, C., Madanshetty, S., & Roy, R. (1988). Thresholds for cavitation produced in water by pulsed ultrasound. *Ultrasonics*, 26(5), 280–285.
98. Chen, H. (2011). *ultra-high speed optical imaging of ultrasound-activated microbubbles in mesenteric microvessels*. Washington, DC: University of Washington.
99. Philipp, A., Delius, M., Scheffczyk, C., Vogel, A., & Lauterborn, W. (1993). Interaction of lithotripter-generated shock waves with air bubbles. *The Journal of the Acoustical Society of America*, 93(5), 2496–2509. <https://doi.org/10.1121/1.406853>.
100. Zhong, P., Cioanta, I., Cocks, F. H., & Preminger, G. M. (1997). Inertial cavitation and associated acoustic emission produced during electrohydraulic shock wave lithotripsy. *Journal of the Acoustical Society of America*, 101(5 Pt 1), 2940–2950.
101. Pishchalnikov, Y. A., Sapozhnikov, O. A., Bailey, M. R., Williams, J. C., Jr., Cleveland, R. O., Colonius, T., et al. (2003). Cavitation bubble cluster activity in the breakage of kidney stones by lithotripter shockwaves. *Journal of Endourology*, 17(7), 435–446. <https://doi.org/10.1089/089277903769013568>.
102. Tomita, Y., & Shima, A. (1990). High-speed photographic observations of laser-induced cavitation bubbles in water. *Acta Acustica United with Acustica*, 71(3), 161–171.
103. Ibsen, S., Benchimol, M., & Esener, S. (2013). Fluorescent microscope system to monitor real-time interactions between focused ultrasound, echogenic drug delivery vehicles, and live cell membranes. *Ultrasonics*, 53(1), 178–184. <https://doi.org/10.1016/j.ultras.2012.05.006>.
104. Ibsen, S., Benchimol, M., Simberg, D., Schutt, C., Steiner, J., & Esener, S. (2011). A novel nested liposome drug delivery vehicle capable of ultrasound triggered release of its payload. *Journal of Controlled Release*, 155(3), 358–366. <https://doi.org/10.1016/j.jconrel.2011.06.032>.
105. Chomas, J. E., Dayton, P., May, D., & Ferrara, K. (2001). Threshold of fragmentation for ultrasonic contrast agents. *Journal of Biomedical Optics*, 6(2), 141–150. <https://doi.org/10.1117/1.1352752>.
106. Brahme, A. (2014). *Comprehensive biomedical physics*. Amsterdam: Elsevier Science.
107. Peng, H. H., Wu, C. H., Kang, S. T., Zhang, J. W., Liu, H. L., Chen, W. S., et al. (2017). Real-time monitoring of inertial cavitation effects of microbubbles by using MRI: in vitro experiments. *Magnetic Resonance in Medicine*, 77(1), 102–111.
108. Coakley, W. (1971). Acoustical detection of single cavitation events in a focused field in water at 1 MHz. *The Journal of the Acoustical Society of America*, 49(3B), 792–801.
109. Roy, R. A., Madanshetty, S. I., & Apfel, R. E. (1990). An acoustic backscattering technique for the detection of transient cavitation produced by microsecond pulses of ultrasound. *The Journal of the Acoustical Society of America*, 87(6), 2451–2458. <https://doi.org/10.1121/1.399091>.
110. Madanshetty, S. I., Roy, R. A., & Apfel, R. E. (1991). Acoustic microcavitation: its active and passive acoustic detection. *Journal of the Acoustical Society of America*, 90(3), 1515–1526.
111. Cleveland, R. O., Sapozhnikov, O. A., Bailey, M. R., & Crum, L. A. (2000). A dual passive cavitation detector for localized detection of lithotripsy-induced cavitation in vitro. *Journal of the Acoustical Society of America*, 107(3), 1745–1758.
112. Hockham, N., Coussios, C. C., & Arora, M. (2010). A real-time controller for sustaining thermally relevant acoustic cavitation during ultrasound therapy. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 57(12), 2685–2694. <https://doi.org/10.1109/tuffc.2010.1742>.
113. Salgaonkar, V. A. (2009). *Passive imaging and measurements of acoustic cavitation during ultrasound ablation*. Cincinnati: University of Cincinnati.
114. Gyöngy, M., & Coussios, C.-C. (2010). Passive cavitation mapping for localization and tracking of bubble dynamics. *The Journal of the Acoustical Society of America*, 128(4), 175–180.
115. Haworth, K. J., Mast, T. D., Radhakrishnan, K., Burgess, M. T., Kopechek, J. A., Huang, S. L., et al. (2012). Passive imaging with pulsed ultrasound insonations. *Journal of the Acoustical Society of America*, 132(1), 544–553. <https://doi.org/10.1121/1.4728230>.
116. Delius, M., & Gambihler, S. (1992). Sonographic imaging of extracorporeal shock wave effects in the liver and gallbladder of dogs. *Digestion*, 52(1), 55–60.
117. Coleman, A. J., Choi, M. J., & Saunders, J. E. (1996). Detection of acoustic emission from cavitation in tissue during clinical

- extracorporeal lithotripsy. *Ultrasound in Medicine and Biology*, 22(8), 1079–1087.
118. Coussios, C., Farny, C., Ter Haar, G., & Roy, R. (2007). Role of acoustic cavitation in the delivery and monitoring of cancer treatment by high-intensity focused ultrasound (HIFU). *International Journal of Hyperthermia*, 23(2), 105–120.
 119. Sapozhnikov, O.A., Bailey, M.R., Crum, L.A., Miller, N.A., Cleveland, R.O., Pishchalnikov, Y.A. et al. (2001) Ultrasound-guided localized detection of cavitation during lithotripsy in pig kidney in vivo. Ultrasonics symposium.
 120. Duck, F. A. (1999). Acoustic saturation and output regulation. *Ultrasound in Medicine and Biology*, 25(6), 1009–1018.
 121. Radiation Tiagon-i (2010) Health effects of exposure to ultrasound and infrasound. In *Documents of the health protection agency, radiation, chemical and environmental hazards*.
 122. Arfelli, F., Rigon, L., & Menk, R. (2010). Microbubbles as X-ray scattering contrast agents using analyzer-based imaging. *Physics in Medicine and Biology*, 55(6), 1643.
 123. Millard, T., Endrizzi, M., Rigon, L., Arfelli, F., Menk, R., Owen, J., et al. (2013). Quantification of microbubble concentration through X-ray phase contrast imaging. *Applied Physics Letters*, 103(11), 114105.
 124. Izadifar, Z., Belev, G., Izadifar, M., Izadifar, Z., & Chapman, D. (2014). Visualization of ultrasound induced cavitation bubbles using the synchrotron X-ray analyzer based imaging technique. *Physics in Medicine and Biology*, 59(23), 7541.
 125. Izadifar, Z., Belev, G., Babyn, P., & Chapman, D. (2015). Application of analyzer based X-ray imaging technique for detection of ultrasound induced cavitation bubbles from a physical therapy unit. *Biomedical Engineering Online*, 14(1), 91.
 126. Stride, E. P., & Coussios, C. C. (2010). Cavitation and contrast: the use of bubbles in ultrasound imaging and therapy. *Proceedings of the Institution of Mechanical Engineers, Part H*, 224(2), 171–191.
 127. Health effects of exposure to ultrasound and infrasound. Health Protection Agency 2010. Report No.: RCE-14, Contract No.: 978-0-85951-662-4.
 128. Carmichael, A. J., Mossoba, M. M., Riesz, P., & Christman, C. L. (1986). Free radical production in aqueous solutions exposed to simulated ultrasonic diagnostic conditions. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 33(2), 148–155.
 129. Crum, L., & Fowlkes, J. (1986). Acoustic cavitation generated by microsecond pulses of ultrasound. *Nature*, 319, 52–54.
 130. Coleman, A. J., Saunders, J. E., Crum, L. A., & Dyson, M. (1987). Acoustic cavitation generated by an extracorporeal shockwave lithotripter. *Ultrasound in Medicine and Biology*, 13(2), 69–76.
 131. Bailey, M. R., Blackstock, D. T., Cleveland, R. O., & Crum, L. A. (1998). Comparison of electrohydraulic lithotripters with rigid and pressure-release ellipsoidal reflectors. I. Acoustic fields. *The Journal of the Acoustical Society of America*, 104(4), 2517–2524.
 132. Lifshitz, D. A., Williams Jr, J. C., Sturtevant, B., Connors, B. A., Evan, A. P., & McAteer, J. A. (1997). Quantitation of shock wave cavitation damage in vitro. *Ultrasound in Medicine & Biology*, 23(3), 461–471. [https://doi.org/10.1016/S0301-5629\(96\)00223-2](https://doi.org/10.1016/S0301-5629(96)00223-2).
 133. Zijlstra, A., Janssens, T., Wostyn, K., Versluis, M., Mertens, P. W., & Lohse, D. (Eds.). (2009). High speed imaging of 1 MHz driven microbubbles in contact with a rigid wall. In *Solid state phenomena*. Trans Tech Publ.
 134. Vignon, F., Shi, W. T., Powers, J. E., Everbach, E. C., Liu, J., Gao, S., et al. (2013). Microbubble cavitation imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 60(4), 661–670.